How much productivity do we need to reduce the workweek?

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Abstract

Brazil proposes to reduce the formal workweek from 44 to 36 hours (PEC 8/2025). An important strand of the Brazilian debate presumes the cut can be absorbed by gains in productivity. The quantitative question is how large that gain would have to be. This paper computes the one-off TFP gain that would leave aggregate output unchanged in the short run — the output-neutrality benchmark A_{req} — using a structural partial-equilibrium model with a CES aggregator over formal and informal effective labor, calibrated to Brazilian microdata. The required gain is $A_{\text{req}} \approx 6.6\%$ under the preferred specification, with a conservative upper bound a couple of percentage points higher. The figure is high relative to the post-1990 Brazilian TFP series, in which productivity fell on average and rose only modestly in the best post-Real-Plan decade. Aggregate informality rises by 1.6–1.9 pp, but the dominant channel of the output cost is the mechanical reduction in hours rather than reallocation. Welfare peaks inside the model near a cap of 42 hours and turns negative below 40 hours, with small firms and the most exposed sectors bearing disproportionate costs. The results favor evaluating a phased reduction with an intermediate stop rather than a single-step cut to 36 hours.

Keywords: working hours, informality, total factor productivity, CES aggregator, Brazil, welfare

JEL: E24, J22, J46, O17, O47

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1 Introduction

An important strand of the Brazilian debate over PEC 8/2025 presumes that the reduction in working hours can be absorbed by gains in productivity. The quantitative question is how large that gain would have to be. This paper computes the one-off TFP gain that would leave aggregate output unchanged in the short run when the statutory workweek falls from 44 to 36 hours. We call this gain the output-neutrality benchmark and denote it A_{req}. The exercise does not assume that productivity will not respond to the reform. It calculates the size of the response that would be required to neutralize the mechanical output loss.

The Brazilian setting complicates the calculation. Informality is 37.8% under the narrow PNAD definition and 44.2% under the broad IBGE one, ranging from 41% in services to 72% in agriculture. Existing models of hours reform tend to abstract from informality (Cacciatore et al., 2016) or treat it through a different margin (Ulyssea, 2018). This paper applies that concern to a broad cut of the statutory cap.

The model is a static partial-equilibrium framework calibrated to Brazilian microdata: a CES aggregator over formal and informal effective labor, an hours–productivity function $e(h)$ peaking at 40 hours, two firm-size groups, and GHH preferences for welfare. The substitution elasticity is pinned to the Brazilian formal–informal wage premium of Meghir et al. (2015). Section 2 reports the calibration in detail.

The central object is A_{req}. Under the preferred *flat-below* specification (per-hour efficiency $e(h)$ flat for hours below the peak) $A_{\text{req}} = 6.63\%$; under the conservative symmetric specification $A_{\text{req}} = 8.18\%$. The joint (σ, ω, η_I) envelope is [5.62%, 7.48%] under the preferred specification and [6.93%, 9.23%] under the conservative one. At Brazil's best post-1990 decade (PWT `rtfpna`, +0.26% per year, 2003–2013) accumulating A_{req} takes 25–30 years. A stress test that layers a 1–2% post-reform TFP contraction raises the conservative A_{req} to 9.28–10.39%. Aggregate informality rises by +1.6–1.9 pp, but the dominant channel of the output cost is the mechanical reduction in hours rather than reallocation. Welfare peaks near a 42-hour cap at about +0.62% (with +0.61% at 41h essentially indistinguishable) and turns negative below 40h.

The paper makes three contributions. First, it embeds a formal–informal reallocation margin into a structural analysis of hours reform and shows the margin is modest under wage-premium-disciplined parameters. Second, it proposes A_{req} as an output-neutrality benchmark commensurable with the historical TFP record. Third, it provides a wage-premium-conditional interval for σ_{sub} that the reader can reassess.

This paper does not estimate the causal effect of PEC 8/2025. It does not model general equilibrium, wage bargaining, worker selection, price adjustment, or capital deepening as central channels. The objective is narrower: to compute the TFP gain that would render the reform output-neutral under parameters disciplined by Brazilian data.

Evidence from Portugal 1996 (Asai et al., 2024) and Brazil 1988 (Gonzaga et al., 2003) enters as a directional plausibility check, not as an identified lower bound for Brazil's effective A_{req}.

The message is simple. A reduction in working hours can raise welfare inside the model, but the cut to 36 hours requires a productivity gain that is large by recent Brazilian standards.

Section 2 presents facts, model, and calibration. Section 3 reports the fit and external plausibility checks. Section 4 reports the counterfactual, the TFP benchmark, and the stress test. Section 5 draws implications for reform design.

2 Brazilian facts and model

Four Brazilian facts motivate the model and discipline the calibration. First, formal hours are concentrated at the statutory maximum. DIEESE (2024) reports 8.5% of formal workers at 36 hours, 26.9% at 40 hours, and 64.6% at or near 44 hours, with an average of 42.2 hours per week. PNAD Contínua Q4 2024 habitual hours give a flatter distribution with an average of 41.2. The majority of formal workers operate at or above 40 hours, the margin a 36-hour cap would bind.

Second, informality is large and dispersed. The national rate is 37.8% (narrow) or 44.2% (broad). The sectoral range is 71.8% in agriculture, 44.4% in industry, and 41.2% in services (Table 1).

Third, the formal–informal wage premium is positive and moderate. Meghir et al. (2015) document a conditional wage premium $R \in [1.15, 1.55]$, the ratio of mean formal hourly wages to mean informal hourly wages among otherwise comparable workers. The calibration below uses this premium to pin down the CES substitution elasticity σ_{sub}.

Fourth, Brazilian TFP has been stagnant or declining throughout the post-1990 period. The Penn World Table 10.01 `rtfpna` series shows an annualized decline of -0.6% per year over 1990–2019 and growth of only $+0.26\%$ per year in the best post-Real Plan decade (2003–2013). The Brazilian TFP benchmark in Section 4 uses this series as the anchor for A_{req}.

These four facts point to a minimal model with three margins: concentration of formal workers near the statutory cap, substitution between formal and informal effective labor disciplined by the observed wage premium, and per-hour productivity that varies with the workweek. The model below combines these three margins with two firm-size groups for differential incidence and GHH preferences for welfare evaluation.

A representative firm in group $g \in \{\text{Small}, \text{Large}\}$ (small: at most 49 employees; large: 50 or more) produces output Y_g from capital K_g and effective labor L_g using a Cobb–Douglas technology,

$$Y_g = A \cdot K_g^{\alpha} \cdot L_g^{1-\alpha}, \tag{1}$$

Table 1: Sectoral and national labor-market moments, PNAD Contínua Q4 2024.

Sector	λ_s	Informality (broad)	Avg. hours (formal)	θ_{36}	θ_{40}	θ_{44}
Agriculture	0.076	71.8%	44.1	0.101	0.269	0.630
Industry	0.205	44.4%	42.5	0.048	0.417	0.535
Services	0.720	41.2%	40.8	0.171	0.410	0.419
National	1.000	44.2%	41.2	0.143	0.406	0.451

Notes: λ_s is the sector's share of national employment. Informality uses the broad IBGE definition (VD4009 combined with V4017 CNPJ indicator). "Avg. hours" is the mean weekly hours of *formal* workers (V4039, habitual). θ_{36}, θ_{40}, θ_{44} are the shares of formal workers in each contracted-hours bin. *Source:* PNAD Contínua Q4 2024, IBGE microdata, survey-weighted, $N = 209{,}311$.

where A is aggregate productivity and α is the capital share of income. Effective labor L_g combines formal (F) and informal (I) workers through a CES aggregator,

$$L_g = \left[\omega L_{F,g}^{\rho} + (1-\omega) L_{I,g}^{\rho}\right]^{1/\rho}, \quad \rho = (\sigma_{\text{sub}} - 1)/\sigma_{\text{sub}}, \tag{2}$$

where ω is the CES weight on formal labor and σ_{sub} is the elasticity of substitution between formal and informal labor. When $\sigma_{\text{sub}} < 1$ the two inputs are complements, and raising the cost of formal labor does not trigger a large reallocation toward the informal margin. The two labor inputs are

$$L_{F,g} = N_{F,g} \sum_b \theta_b h_b e(h_b), \qquad L_{I,g} = \eta_I N_{I,g} h_I e(h_I),$$

where $N_{F,g}$ and $N_{I,g}$ are the headcounts of formal and informal workers in group g (with $N_{I,g} \equiv N_g - N_{F,g}$ and total group employment N_g fixed), h_b indexes the three contracted-hours bins $b \in \{36, 40, 44\}$ with shares θ_b among formal workers, $h_I = 44$ is weekly hours for informal workers, and η_I is the per-head productivity of an informal worker relative to a formal worker. Per-hour efficiency follows $e(h) = \exp\{-\kappa(h-h^*)^2\}$ above the peak $h^* = 40$, where κ is the curvature of the fatigue penalty. Below the peak the paper considers two regimes. The preferred flat-below specification sets $e(h) = 1$ for $h \leq h^*$. The conservative symmetric specification keeps the same quadratic form below h^*. The flat-below specification is informed by Pencavel (2015), who documents the fatigue leg above the peak but does not characterize efficiency below it. The symmetric specification serves as a conservative upper bound on the required TFP gain.

Each firm chooses the number of formal workers $N_{F,g}$ to maximize profit net of three cost wedges,

$$Y_g - \tau_g N_{F,g} - \tfrac{1}{2}\pi_m N_{I,g}^2 - \tfrac{1}{2}\gamma_{F,g}(N_{F,g} - \bar{N}_{F,g})^2, \tag{3}$$

where τ_g is a linear per-worker cost of formality (CLT/FGTS obligations and size-dependent enforcement), π_m is a convex cost of informality (fines, reputational risk, and limited access to formal inputs), and $\gamma_{F,g}$ is a quadratic adjustment cost penalizing deviations from the pre-reform formal headcount $\bar{N}_{F,g}$. The wedge τ_g is calibrated to match group-specific informality and plays the role of a size-dependent distortion in the spirit of Hsieh and Klenow (2009) and Restuccia and Rogerson (2008), with endogenous-formality micro-foundations in Dix-Carneiro et al. (2026). Cutting the statutory cap from $h_0 = 44$ hours to h_1 reduces effective labor through both the hours bins h_b and the efficiency function $e(h_b)$, and widens the wedge between the marginal product of formal labor and its private cost.

The object of interest is the TFP multiplier that restores aggregate output,

$$A_{\text{req}}: \quad \sum_g Y_g(A\,(1 + A_{\text{req}}), K_g, h_1) = \sum_g Y_g(A, K_g, h_0). \tag{4}$$

A_{req} is the one-off percentage gain in aggregate productivity A that exactly offsets the short-run output loss from moving the cap from h_0 to h_1. It is an accounting benchmark to compare with Brazil's historical TFP record, not a predicted growth rate.

For welfare, preferences are GHH (Greenwood et al., 1988) with $\sigma_{\text{ghh}} = 1$ and Frisch elasticity $1/\nu = 0.5$. The compensating variation is

$$\Delta\text{CV} = \frac{C_1 - \psi h_1^{1+\nu}/(1+\nu)}{C_0 - \psi h_0^{1+\nu}/(1+\nu)} - 1. \tag{5}$$

The numerator and denominator are the post- and pre-reform consumption-equivalents of welfare under GHH preferences, computed as consumption net of the monetized labor disutility $\psi h^{1+\nu}/(1+\nu)$. ΔCV is the percentage change in welfare.

The model is static and partial-equilibrium. Capital is predetermined, total employment N is fixed, and general-equilibrium feedbacks (wage-price, monetary, fiscal, cross-sector) are absent. The online appendix enumerates the omitted channels and bounds their aggregate effect. A textbook user-cost elasticity $\varepsilon_K \in [0.2, 0.4]$ (Caballero, 1999) with capital share $\alpha = 0.35$ implies

$$\Delta A_{\text{req}}^K \approx -\alpha \cdot \varepsilon_K \cdot |\Delta L_F/L| \in [-1.0, -1.8] \text{ pp}, \tag{6}$$

where ΔA_{req}^K is the downward correction to the short-run A_{req} once the endogenous response of the capital stock is factored in. The medium-run analog of the headline thus falls by 1–2 pp over a 3–5 year horizon.

Table 2 lists the parameters under three headings. External parameters come from PWT, DIEESE, and PNAD Contínua. Moment-matched parameters (τ_g, π_m, κ, ψ) are calibrated internally. Author-chosen parameters (η_I, e_q, h^*, h_I, γ_F^g) are set inside ranges consistent with the work-hours evidence of Pencavel (2015) and Collewet and Sauermann (2017), the informal-productivity literature (La Porta and Shleifer, 2014), and the CLT/FGTS compliance-cost literature. Cross-country hours evidence in Bick et al. (2018)

provides directional support for declining marginal product of hours but does not identify a peak. The online appendix reports the source and sensitivity for each parameter.

Table 2: Parameter values by evidential basis.

Symbol	Description	Value	Source
External			
α	Capital share	0.35	PWT 10.01 + Gollin (2002)
ω	CES formal weight	0.622	1− narrow informality (PNAD Contínua 2025)
θ_b	Hours-bin weights (formal)	$(0.085, 0.269, 0.646)$	DIEESE 2024
ν	Inverse Frisch elasticity	2.0	Standard (Frisch = 0.5)
Moment-matched			
σ_{sub}	CES substitution elasticity	**1.326** $[1.116, 1.469]$	Wage premium (Meghir et al., 2015)
τ_S, τ_L	Formal wedges (small, large)	$2.609, 0.004$	Bisection to group informality
π_m^S, π_m^L	Informality convex cost	$0.000, 42.11$	Bisection
κ	Efficiency curvature	2.11×10^{-3}	Derived from e_q at h_{ref}
ψ	GHH disutility weight	5.74×10^{-5}	Baseline FOC
Author-chosen (within ranges consistent with evidence)			
η_I	Informal relative efficiency	0.40	La Porta and Shleifer (2014) informal/formal wage ratio midpoint ($\approx 1/3$ to $1/2$)
e_q	Hours–output elasticity	0.60	Pencavel (2015) range
h^*	Peak-efficiency hours	40	Pencavel (2015)
h_I	Informal hours	44	No statutory cap in informal sector
γ_F^S, γ_F^L	Adjustment costs	$0.12, 0.03$	RAIS + CLT/FGTS

The CES substitution elasticity σ_{sub} is disciplined by the wage premium. From the CES first-order condition, the formal–informal wage premium R is a monotone function of σ_{sub}. Plugging a widened window around Meghir et al.'s estimates, $R \in [1.15, 1.55]$, into the CES FOC at the recalibrated $(\omega, \eta_I) = (0.622, 0.40)$ yields

$$R \in [1.15, 1.55] \implies \sigma_{\text{sub}} \in [1.116, 1.469], \quad A_{\text{req}} \in [7.50\%, 8.58\%] \text{ (conservative, midbox).} \tag{7}$$

The central value $\sigma = 1.326$ corresponds to $R \approx 1.40$, the midpoint of the MNR window.[1]

[1] The window $R \in [1.15, 1.55]$ spans the full range of MNR's reported estimates across sub-populations and alternative specifications (Tables 3 and 10). MNR's R is a structural premium estimated inside their search-and-matching model; inverting it via the CES first-order condition assumes the relative wage MNR identifies between formal and informal workers coincides with the relative marginal product of formal vs. informal effective labor inside the CES aggregator. A reduced-form formal-vs-informal hourly wage ratio in PNAD Contínua 2024–2025, conditional on standard occupation, education, and experience controls, falls inside the MNR window and supports this identification. The mapping from R to σ_{sub} depends on (ω, η_I). The anchors $\omega = 0.622$ (1 minus the narrow PNAD 2025 informality rate) and $\eta_I = 0.40$ (midpoint of the La Porta–Shleifer informal/formal wage ratio range 1/3 to 1/2) are theoretically consistent: under competitive wage-taking the CES aggregator identifies η_I through the

This wage-premium-disciplined calibration supersedes earlier author-chosen values of σ_{sub} used in preliminary rounds of the paper. The shift carries one qualitative implication for the policy story. Under $\sigma_{\text{sub}} < 1$ formal and informal labor are complements, and a tighter formal cap pushes inputs together rather than apart; the informality response is muted or even negative. Under $\sigma_{\text{sub}} > 1$, as disciplined here by the MNR window, they are imperfect substitutes, and the cap pushes marginal formal workers toward informality. The aggregate informality response in the counterfactual rises rather than falls. The fourth policy message in Section 5 should be read in this light: the reallocation channel is material, not absorbed by intensive adjustment alone. Letting ω and η_I also vary over their data-consistent ranges ($\omega \in [0.58, 0.66]$, $\eta_I \in [0.33, 0.50]$), the joint (σ, ω, η_I)-box envelope under the conservative specification is $[6.93\%, 9.23\%]$ at the eight corners. Under the preferred flat-below specification the central A_{req} is 6.63% and the envelope is $[5.62\%, 7.48\%]$. The online appendix reports the full grids. The small reallocation elasticity documented by Derenoncourt et al. (2025) supports a modest intra-firm σ and is distinct from the high substitution elasticities between worker types reported in refugee studies. The online appendix discusses this distinction.

Rebuilding the calibration on PNAD habitual hours ($\theta = (0.143, 0.406, 0.451)$, 44.2% informality) yields $A_{\text{req}} = 7.80\%$ under the conservative specification. The baseline uses DIEESE contracted hours because the statutory cap is a contracted-hours concept.

3 Fit and external plausibility

The role of this section is narrow. The model matches the calibration targets by construction; the external benchmarks below are read as directional plausibility checks, not as causal validation of the counterfactual. Magnitudes from the Portuguese 1996 reform and the Brazilian 1988 reform should be compared in sign and order, not in level. Table 3 reports the calibration targets, the domestic alternative-measurement check, and the external directional benchmarks together. Panel A targets are matched by construction. Panel B1 evaluates the model against the PNAD-habitual hours measurement, which the calibration does not target. The implied gap of 1.0 h/week on mean formal hours and 5.8 pp on the share above 36h reflects the contracted-versus-habitual measurement distinction the baseline commits to. To verify that the Panel B1 moments are not algebraic consequences of the calibration, the replication package perturbs the formal hours distribution θ within a ± 5 pp envelope around the DIEESE 2024 weights and re-runs the calibration: the fitted Panel B1 moments shift by 0.4–0.6 h/week and 1–2 pp, with the sign of the gap preserved but its magnitude responsive to the perturbation. The non-targeted moments therefore provide an informative check outside the calibration constraints; they are not

marginal-product (wage) ratio rather than through value-added per employee, which is contaminated by capital-per-worker differences. The online appendix documents the derivation.

mechanical reflections of θ. Panel B2 reports the Portuguese 1996 reform at the common $44 \to 40$h cap and the Brazilian 1988 reform at its $48 \to 44$h cap. Signs match. Aggregate output falls and hourly productivity rises in both data and model. Magnitudes do not match. The firm-level Portuguese estimate of +4.4% in hourly productivity is about 2.6 points above the aggregate model prediction, a gap consistent with worker selection, work intensification, and sectoral capital deepening that the model does not encode. The Brazilian 1988 evidence points the same way with a slightly larger empirical hourly-wage response. Panel B2 should be read as a directional plausibility benchmark rather than an identified Brazilian lower bound for the effective A_{req}. Portugal and Brazil 1988 differ from the PEC 8/2025 environment in informality, institutional setting, and reform magnitude.

Table 3: Calibration fit: targets and checks outside calibration.

Moment	Data	Model	Gap
Panel A: Calibration targets (matched by construction)			
Informality rate, small firms ($\leq$ 49 empl.)	0.500	0.500	0.000
Informality rate, large firms	0.200	0.200	0.000
Formal employment share, small firms	0.590	0.590	0.000
Formal employment share, large firms	0.410	0.410	0.000
Aggregate informality, narrow PNAD	0.378	0.377	0.001
Wage premium R, MNR (2015) central target	1.400	1.400	0.000
Panel B1: Domestic checks outside calibration			
Average weekly formal hours (DIEESE vs PNAD habitual)	41.23	42.24	1.01 h/week
Share of formal workers above 36h (DIEESE vs PNAD, %)	85.7	91.5	5.80 pp
Aggregate weekly hours (formal + informal)	39.13	42.91	3.78 h/week
Panel B2: External directional benchmarks			
ΔY at $44 \to 40$h cap, Portugal 1996 (firm-level)	−3.20	−2.00	1.20 pp
$\Delta Y/h$ at $44 \to 40$h cap, Portugal 1996 (%)	+4.40	+1.77	2.63 pp
$\Delta W/h$ at $48 \to 44$h, Brazil 1988 (%, GMC 2003)	+7.00	+1.77	5.23 pp

Panel B1 rows 1–2 exploit the DIEESE-contracted vs PNAD-habitual measurement distinction. Row 3 reports aggregate weekly hours under the $h_I = 44$ assumption, sensitivity-checked in the online appendix (A_{req} shifts by < 0.2 pp for $h_I \in [38, 46]$). Panel B2 compares the model's aggregate prediction at the $44 \to 40$h cap to the Portuguese firm-level estimate (Asai et al., 2024) and to Brazil's 1988 reform of comparable cap-cut magnitude (48→44h, ∼8%) as documented by Gonzaga et al. (2003). Both external benchmarks suggest the model is conservative on the hourly-productivity response. The model's $e(h)$ captures fatigue and not worker selection, work intensification, or sectoral capital deepening. Sources: PNAD Contínua Q4 2024, RAIS 2022, Meghir et al. (2015), Asai et al. (2024), Gonzaga et al. (2003).

4 The productivity arithmetic of the 36-hour cap

Table 4 reports the aggregate effects of the reform under both $e(h)$ regimes, holding total employment fixed. Under the preferred flat-below specification, output falls by $\Delta Y = -6.60\%$, aggregate welfare by $\Delta \text{CV} = -1.76\%$, and the required TFP gain is $A_{\text{req}} = 6.63\%$. Under the conservative symmetric specification, $\Delta Y = -8.02\%$, $\Delta \text{CV} = -3.63\%$, and $A_{\text{req}} = 8.18\%$. Aggregate informality rises by +1.6–1.9 pp under both specifications. With moderate informal relative efficiency ($\eta_I = 0.40$, identified from the observed wage ratio) and mildly substitutable formal–informal labor ($\sigma > 1$), firms shift marginal workers toward informality when the cap tightens instead of absorbing the hours loss intensively. Varying the three load-bearing parameters (σ, ω, η_I) at the corners of their data-consistent ranges gives a joint envelope of $A_{\text{req}} \in [5.62\%, 7.48\%]$ under the preferred specification and [6.93%, 9.23%] under the conservative one. The preferred envelope lies strictly below the conservative one because the flat-below fatigue channel contributes positively to per-hour productivity, which the symmetric specification does not deliver. The capital-adjustment bound of Equation (6) lowers the effective A_{req} by 1–1.8 pp at textbook elasticities over a 3–5 year horizon, placing the medium-run analog at roughly 4.8%–5.6% under the preferred specification and 6.4%–7.2% under the conservative one.

Table 4: Main results: 44 → 36h reform.

	Flat-below (preferred)	Symmetric (conservative)
Positive predictions (no TFP compensation)		
Output, ΔY	−6.60%	−8.02%
Welfare (GHH), ΔCV	−1.76%	−3.63%
Informality, ΔInf	+1.57 pp	+1.92 pp
Output per hour, $\Delta Y/h$	+2.39%	+0.75%
Benchmark: TFP gain that restores output		
A_{req}, central	6.63%	8.18%
A_{req}, joint envelope	[5.62, 7.48]%	[6.93, 9.23]%

The joint envelope spans the eight corners of the disciplined (σ, ω, η_I) box. The preferred envelope lies strictly below the conservative one because the flat-below specification has a positive fatigue channel that the symmetric one does not.

The curve in Figure 1 is convex. The first hours of reduction cost little in TFP terms because they move formal workers toward the efficiency peak at $h^* = 40$. Subsequent reductions grow more expensive as the cap overshoots the optimum. The output loss decomposes into a mechanical hours channel (approximately −14.8%, identical across specifications) and a fatigue channel that switches sign between specifications, from small-positive under flat-below to small-negative under symmetric. The CES reallocation residual is positive and moderate. The condition $\sigma > 1$ makes formal and informal labor imperfect substitutes, and the moderate $\eta_I = 0.40$ dampens but does not eliminate the reallocation cost when informality rises. The online appendix reports the full decomposition figure.

We anchor A_{req} against post-1990 Brazilian TFP history using the PWT 10.01 `rtfpna` series (Feenstra et al., 2015). Figure 2 uses the preferred baseline $A_{\text{req}} = 6.63\%$. Brazilian TFP fell by −0.60% per year over 1990–2019 and rose by only +0.26% per year in the best decade (2003–2013). At the best-decade rate, accumulating the preferred A_{req} takes approximately 25 years and the conservative one takes 30 years. The +2.3% per year of

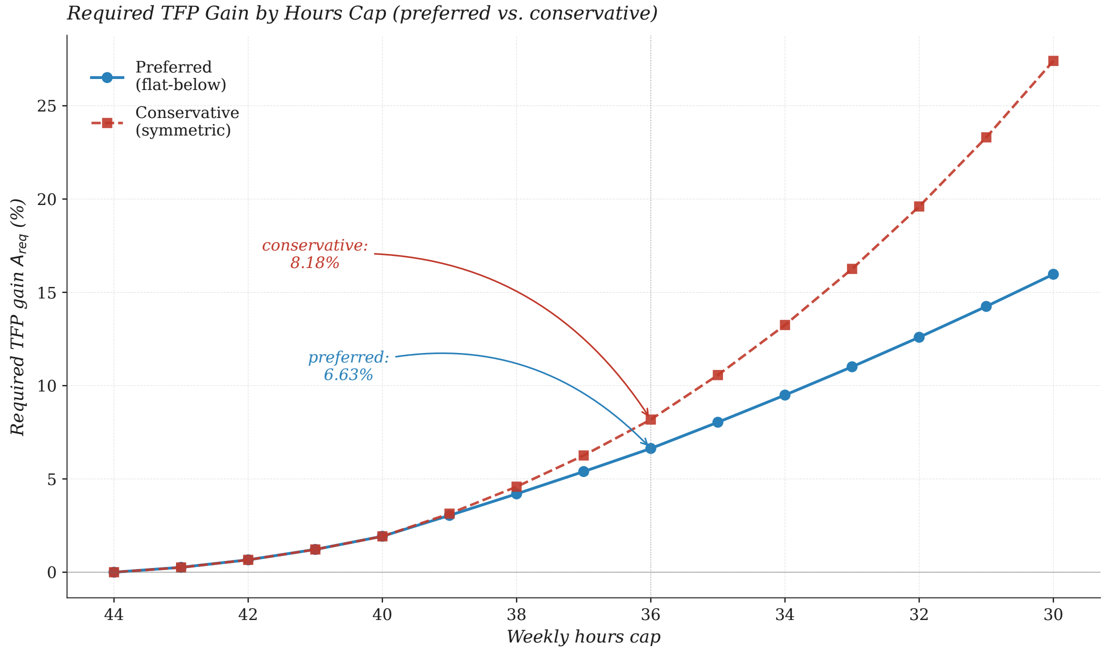

Figure 1: Required TFP gain A_{req} as a function of the weekly hours cap.
Notes: The preferred flat-below schedule yields 6.63% at the proposed 36h cap. The conservative symmetric schedule yields 8.18%. The two schedules coincide above $h^* = 40$ and diverge below it, reflecting different assumptions about $e(h)$ below the efficiency peak.

1954–1980 reflects the Brazilian economic miracle and is not a realistic reference point. The online appendix reports the full table. The preferred benchmark is high relative to any horizon in the post-1990 Brazilian series.

A pessimistic stress test sets post-reform TFP to $A_{\text{post}} = A(1 - \delta)$ for $\delta \in [0\%, 2\%]$, capturing disruption to learning-by-doing and hysteresis in the informal sector. Table 5 reports the result under the conservative specification. A TFP contraction of 1% raises A_{req} from 8.18% to 9.28% and the welfare loss from 3.63% to 4.89%. A 2% contraction, a magnitude observed in 2015–2016, raises A_{req} to 10.39%. The stress test does not predict a particular δ. It shows that the central prediction does not collapse under moderate post-reform TFP deterioration.

The aggregate compensating variation at the $44 \to 36$h reform is -1.76% under the preferred flat-below specification and -3.63% under the conservative symmetric one, both computed from Equation (5). Figure 3 plots the full schedule. Welfare peaks at approximately $+0.62\%$ near a 42h cap under both specifications, with $+0.61\%$ at 41h essentially indistinguishable. The leisure gain outweighs the consumption loss in this range. The two schedules coincide between 44h and 40h and diverge below 40h. The coincidence reflects a structural feature. At any cap in $[40, 44]$ the binding workers come from the 44h bin, and their per-hour efficiency rises toward the peak under both specifications (from $e(44) = 0.967$ toward $e(40) = 1$). The divergence below 40h reflects the difference in specification. The flat-below case keeps $e(h) = 1$ for $h \le 40$, while the symmetric case

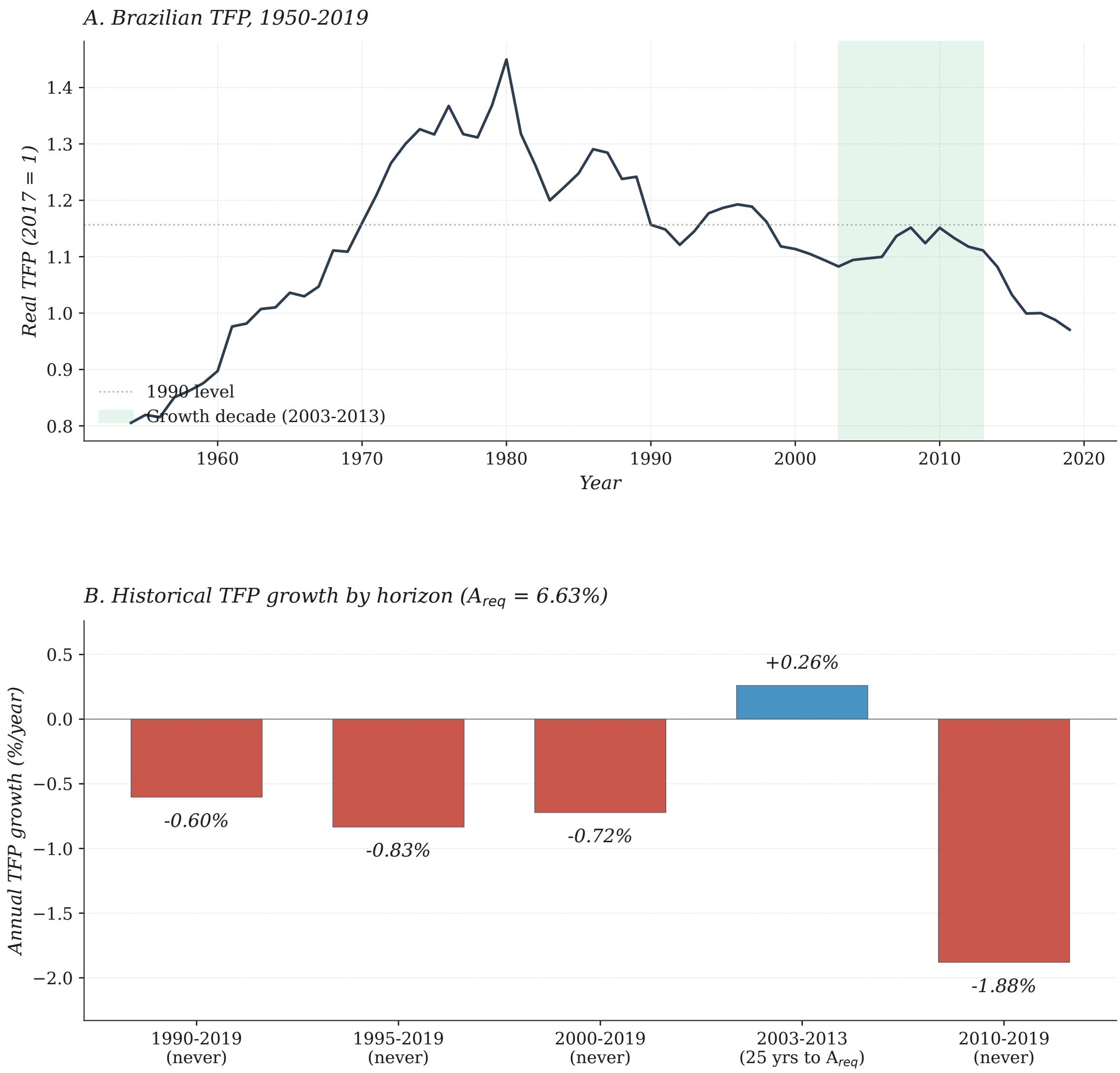

Figure 2: Brazilian TFP and the required one-off productivity gain.

Notes: Panel A plots the PWT `rtfpna` series (real TFP at constant national prices) for Brazil, 1954–2019. Panel B shows annualized TFP growth rates by horizon. Labels indicate the years required to accumulate the preferred baseline $A_{\text{req}} = 6.63\%$ at each historical rate ("never" when the rate is non-positive). *Source:* Penn World Table 10.01 (Feenstra et al., 2015).

Table 5: Pessimistic stress test: reform with post-reform TFP shock.

Post-reform TFP shock δ (%)	A_{req} (%)	ΔY (%)	ΔInf (pp)	ΔCV (%)
0.0 (baseline)	+8.18	−8.02	+1.917	−3.63
0.5	+8.73	−8.51	+1.980	−4.26
1.0	+9.28	−9.00	+2.044	−4.89
1.5	+9.83	−9.50	+2.122	−5.54
2.0	+10.39	−9.99	+2.185	−6.17

δ represents a post-reform contraction of TFP beyond the mechanical effect of the hours cap. The $\delta = 0$ row reproduces the main result of Table 4. Other rows show how outcomes deteriorate if the reform itself depresses TFP by δ.

assigns an efficiency penalty that accumulates as the cap moves further below the peak. Welfare turns negative below 40h in both specifications. A moderate reduction from 44 to 41–42 hours raises welfare inside the model. The full 44 → 36 reform does not.

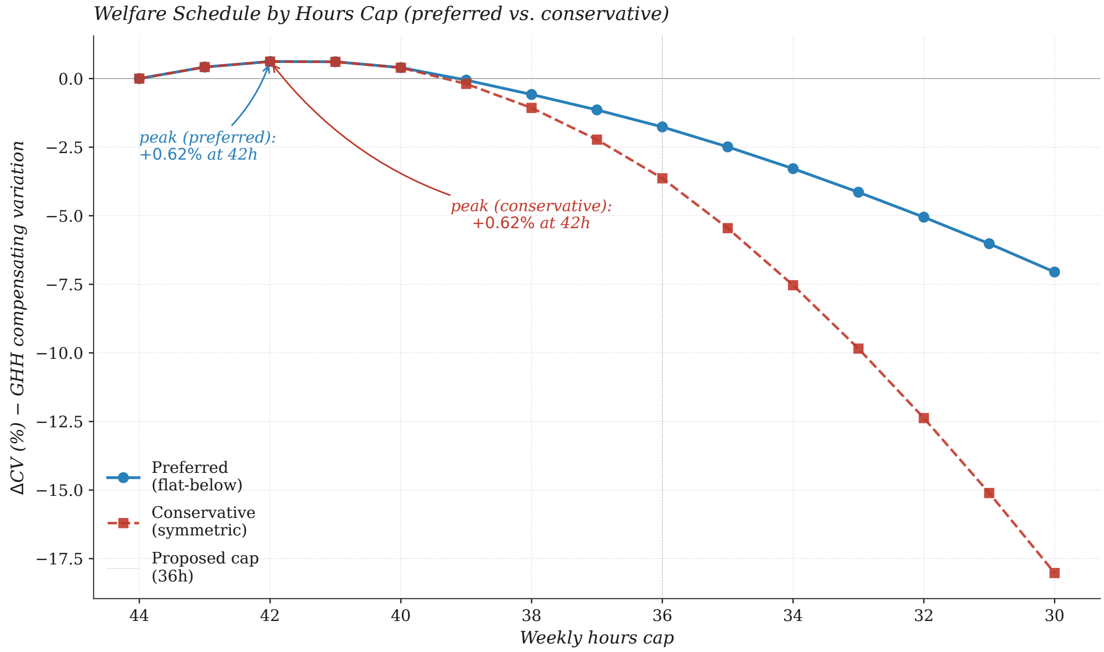

Figure 3: Welfare schedule by hours cap under both calibrations.

Notes: Each line plots aggregate welfare ΔCV from Equation (5) at successively lower hours caps, relative to the 44h baseline, under the preferred flat-below and conservative symmetric $e(h)$ specifications. ΔCV peaks near 42h at about +0.62% and turns negative below 40h. At the proposed 36h cap, welfare falls by 1.76% under the preferred specification and by 3.63% under the conservative one.

A TFP gain of approximately 1.4% at the 36h cap is enough to make the reform welfare-neutral under the preferred specification. The conservative threshold is 2.9%. Both thresholds lie below the output-neutral A_{req} because leisure partially compensates the consumption loss. Welfare-neutral and output-neutral TFP differ by construction. The online appendix reports a type-level decomposition and reconciles the GHH-nonlinearity gap between type-level and representative-agent CV.

The welfare schedule does not imply opposition to reductions in working hours. It indicates that, under the model's calibration, the welfare-improving margin lies near 41–42 hours, not at 36. The schedule also separates three objects that the public debate often blurs: the output loss, which the reform mechanically delivers under fixed productivity; the productivity gain that would offset that loss, summarized by A_{req}; and the welfare effect, which depends on the value workers place on extra leisure. The three objects move differently with the cap, and the welfare-improving region is narrower than the output-neutral one.

Adjustment costs and informality differ across firm sizes and sectors, producing heterogeneous incidence of the reform. Small firms ($\leq$ 49 employees) require a larger TFP gain than large firms, reflecting their higher formalization wedges and baseline informality. Applying the national framework separately to three broad sectors (agriculture, industry, services) with sector-specific PNAD parameters delivers a per-sector A_{req} of 7.60% in agriculture, 8.05% in industry, and 6.91% in services under the conservative specification, with aggregate informality rising in every sector (agriculture +2.88 pp, industry +2.73 pp, services +2.10 pp). Industry's A_{req} slightly exceeds agriculture's because the moderate $\eta_I = 0.40$ combined with industry's larger formal base dampens the mechanical gap between the two sectors. Services account for most of the aggregate output loss through their output share, not through a larger per-worker effect. The online appendix reports the full decomposition and the parameter sensitivities.

5 Implications for reform design

The paper quantifies a trade-off the Brazilian policy debate has so far left implicit. Cutting the statutory workweek from 44 to 36 hours affects the majority of Brazilian formal workers and reduces the labor input of the economy by about 14% mechanically. Whether the economy absorbs this cut without an output loss depends on the size of the simultaneous productivity gain. The paper puts a number on that gain. The aggregate output loss is offset by a one-off productivity gain of 6.63% (preferred specification) to 8.18% (conservative upper bound). Both figures lie above any horizon-equivalent gain in the post-1990 Brazilian productivity record. In the best decade after the Real Plan, productivity rose by about a quarter of a percent per year. Matching what the reform requires at that rate would take twenty-five to thirty years.

Four messages follow from the model.

First, the 36-hour proposal imposes a high productivity requirement by the standards of the post-1990 Brazilian record. The reform requires a one-off productivity gain that exceeds the cumulative gain produced in the best post-Real-Plan decade. Without that gain, the model predicts a short-run output loss of 6.6 to 8.0% and an aggregate welfare loss of 1.8 to 3.6%. The productivity requirement is a benchmark, not a forecast of what TFP will do. In the absence of an unusual productivity acceleration, a meaningful share of this loss would be expected to materialize.

Second, a moderate reduction in the cap is welfare-improving inside the model. Cutting the workweek from 44 to 41–42 hours raises aggregate welfare because the value workers place on extra leisure exceeds the consumption they give up. The welfare-improving region spans 41–43 hours; below 40h welfare turns negative. The welfare-maximizing cap inside the model is near 42 hours, with 41h essentially indistinguishable; the proposed 36-hour cap lies below the welfare-maximizing cap of the model. This interior maximum is a feature of the model, not a policy recommendation.

Third, the cost of the reform is concentrated. Small firms, which employ most formal workers in Brazil, bear the largest per-worker productivity requirement. Industry and agriculture carry the largest per-sector requirements; informality rises in all three sectors rather than falling. Services account for the bulk of the aggregate output loss through their share in employment and output, not through a larger per-worker effect. Reform design that ignores this concentration will understate the adjustment burden.

Fourth, informality rises but is not the dominant channel of the cost. The calibrated model, disciplined by the observed formal–informal wage premium, finds an aggregate informality increase of 1.6–1.9 pp. This rise is material for welfare accounting and small relative to the roughly 15% mechanical hours contraction. The output cost comes primarily from fewer hours worked. The reform also pushes additional workers into informality, weakening its own formalization goals.

The interpretation requires separating output, required TFP, and welfare. A moderate reduction can lower output, raise welfare, and still require a positive but small TFP gain, which is what the model delivers between 44 and 42 hours.

These conclusions come with limits. The model is short-run and partial-equilibrium. Endogenous capital adjustment over a three-to-five-year horizon lowers the required productivity gain by about one to two percentage points. Total employment is held fixed, which shuts down work-sharing. The empirical evidence on work-sharing from Germany (Hunt, 1996) and Brazil's 1988 constitutional reform (Gonzaga et al., 2003) suggests this restriction is compatible with the available evidence but not innocuous. The Portuguese 1996 experience (Asai et al., 2024) documents firm-level productivity gains larger than the model produces, which suggests the model is conservative on the hourly-productivity response.

Subject to these limits, the productivity arithmetic favors a phased path. A reduction with an intermediate stop near 41–42 hours is consistent with the model's welfare schedule, keeps the productivity requirement closer to the gains observed in the best post-Real-Plan decade, and buys time for the capital stock, the informal margin, and firm-level productivity to adjust before the next step. The single-step move to 36 hours places the requirement above any horizon-equivalent gain in the post-1990 Brazilian series. The implication is not that the workweek should remain at 44 hours; it is that the size and the timing of the reduction matter for whether the reform delivers a welfare gain or a welfare loss.

Online appendix. An online appendix containing data sources, partial-equilibrium omissions, calibration details, sensitivity grids, sectoral decomposition, type-level welfare, output decomposition, the historical TFP table, derivations, and the asymmetric-$e(h)$ robustness is available at `[URLpending]`.

Code and data availability. Code and calibration targets to reproduce every numbered table and figure in this paper are available at the author's public repository `https://github.com/vr-rodrigues/jornada-6x1-replication` (to be activated upon publication).

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